WEEKEND CHALLENGE: MAP IT · LIVE DEMO

Cloudcroft Disc Golf

An AR map + scorecard at 8,676 ft - surveyed on foot, mapped in your browser.

Open it live: **bryanchasko.com/cloudcroft-disc-golf/**

Two real courses. One November field survey. Zero invented pins.

Bryan Chasko · 3-minute live demo · every claim in this deck traces to the repo



BEFORE YOU CLICK

The exact words you will see on screen

0:10-0:25

PENDING REFRESH - being smoke-tested

These labels are in the refresh Bryan is smoke-testing now. They are not on the live site yet (verified 2026-11-02). If the refresh has not shipped when you record, use the live words in Appendix A - never

Buttons: Byron Ligon · Cloudcroft Community · Mountaintop · Sky-Cam · First Person · VR · ? (Help)

Help text, verbatim: *"Drag to look around · scroll to zoom in Sky-Cam · in First Person, WASD or arrow keys move, Shift hurries · Mountaintop goes to Community tee box 2 · in a pano, drag left–right."*

In a pano: *"drag left–right to look around" · ‹ › · Back*

Try-it path, verbatim: *"Mountaintop → tee box 2 card → next/prev through the panos → First Person → walk the fairway."*

ALREADY LIVE - safe to promise

bryanchasko.com/cloudcroft-disc-golf/ (HTTP 200)

"Open the live map" → /app/ (HTTP 200)

Walk page buttons today: Orbit · Walk (WASD) · "Stand on the mountaintop — tee box 2"

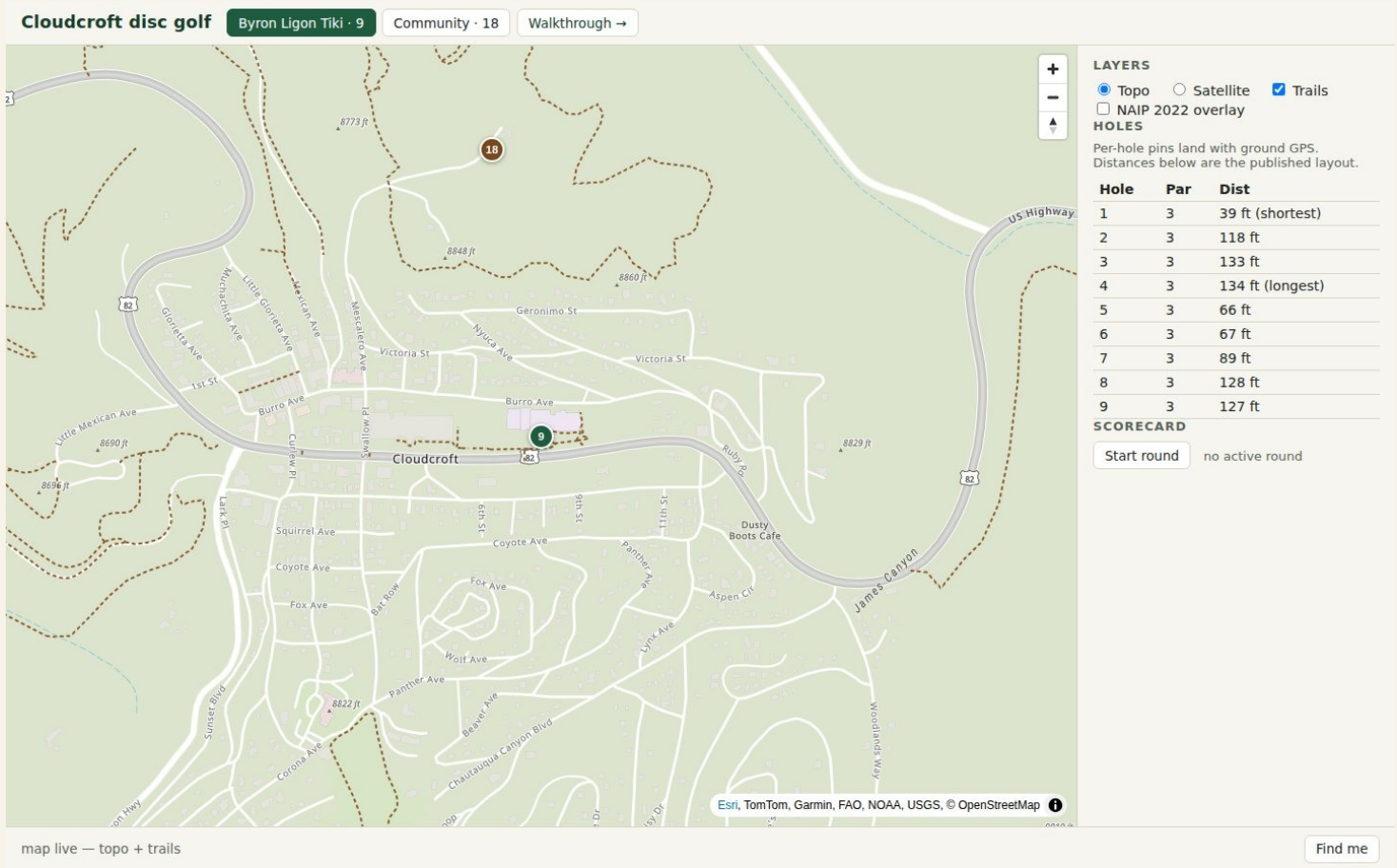
Rule for recording: read the screen, not this deck.

Demo flow for the next 2½ minutes: open the landing page → enter the map → Mountaintop at Community tee box 2 → move around and open field photos + 360° panos → how the survey became the map → the AWS pattern you can reuse.

LIVE FLOW - STEP 1

Open the site. Enter the map.

0:25-0:47



Repo QA capture of the live 2D map (Oct 31) - on camera, show the real browser, not this still.

- 1. Open **bryanchasko.com/cloudcroft-disc-golf/**
- 2. Click **“Open the live map”** - you land in /app/
- Two courses, one toggle: Byron Ligon Tiki · 9 and Community · 18
- Topo / Satellite, Trails, and the NAIP 2022 overlay are layers on the same map
- The hole table and scorecard sit beside the map - no app install,

LIVE NOW - verified HTTP 200

Mountaintop: Community tee box 2

0:47-1:15



- Click **Mountaintop** (live today: “Stand on the mountaintop — tee box 2”)
- You land on the highest surveyed pad: 40-fix GPS average, elevation 2,675.3 m
- This 360° was shot 1.4 m from that average - the view and the coordinate are the same field moment
- **Drag left–right to look around.** That instruction is printed in the pano itself.

Above: the actual tee-box-2 360°, flattened. On camera you stand inside it and drag.
Below: the sign Bryan photographed at that pad.



Hole 2 sign, read on site: 245 ft · par 3 · mando right · fence-line O.B. right

The app card says the same words - sign, survey remark, and map agree.

TEE-2 360°: PENDING REFRESH

Until the refresh ships, the live walk shows this pad's real field photos at tee box 2. Say whichever is on screen.

LIVE FLOW - STEP 3

Move around. Open what the field day actually captured

1:15-1:48







Basket 1 field photo · course kiosk field photo · Pano A (Hole 1 tee area), flattened - in the walk you drag inside it

- **On the 2D map:** choose a point, photo, or panorama - every survey point opens its photos, and panos link onward to "stand here in the 3D walk".
- **In the 3D walk:** Sky-Cam to orbit, First Person + WASD to walk the fairway, ‹ › to step through the six panos (A–F in the pano index).
- **The mapped field record:** 15 Pixel photos · 6 panoramas · 20 SW Maps screenshots · 21 Community points · 6 Tiki points (survey-assets.json counts).

Survey to map: no hand-placed pins

1:48-2:10

1 · WALK

SW Maps on a phone: averaged GPS points + a written remark at every pad and basket

2 · CONVERT

swm2_to_geojson.py turns the .swmz field backup into points.geojson

3 · VALIDATE

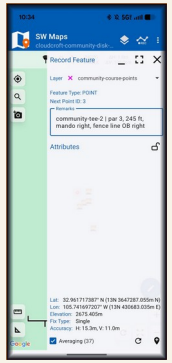
validate_survey.py checks counts, names, bounds, notes - gaps are reported, never silently filled

4 · BUILD

build_holes.py samples the 1 m DEM, cross-checks the tee signs, precomputes plays-like distance

5 · PUBLISH

GeoJSON + web panos + photos ship to the same map you just used - next survey drop needs no code change



The real capture screen for the pad you just stood on: “community-tee-2 | par 3, 245 ft, mando right, fence line OB right” - averaging 37 fixes at that moment, elevation 2,675.405 m.

That remark is now map data. The pipeline is scripts + tests in the repo (dataset/scripts/), so anyone can re-run it on their own course survey.

UNDER THE HOOD - 15 SECONDS

The AWS shape (one CDK stack)

2:10-2:28

VISITOR	AMAZON LOCATION SERVICE	API + DATA	HOSTING + ACCESS
• Browser - no install • MapLibre 2D map • Babylon.js 3D walk	• Topo + imagery maps • Place index (geocoded anchors) • Per-hole geofences · tracker · routes	• API Gateway (HTTP API) • Rust ARM Lambdas: plays-like + scorecard • DynamoDB scorecards	• S3 dataset + web bundle • CloudFront (site distribution) • Cognito guest pool - one scoped role

Plays-like, in one line: horizontal distance plus elevation change over the 1 m DEM - verified live at 364.8 ft + 40 ft climb = 404.8 ft on a synthetic hole (build log, day 0).

One stack owns it all: TypeScript CDK in us-east-1 (infra-cdk/) - Location resources, guest identity pool, both Lambdas, the table, and the API. Zero hand-made resources remain.

Not in this demo: the Android AR companion is a stretch goal and is not scaffolded. Do not promise on-tee AR on camera.

PROVENANCE

Open data actually used - and what we rejected

2:28-2:40

- **USGS 3DEP 1 m DEM** (NM_SouthEast_2018_D19) - one tile covers both courses; anchor samples match the national elevation service to 0.1 ft. This is the elevation source of record.
- **USGS NAIP 2022, 60 cm** - one tile covers both courses; you saw it as the map overlay and the 3D terrain drape.
- **OpenStreetMap** - live Overpass extract: 2,500 elements, 46 paths + 26 footways with real trail names. Neither course was in OSM - this dataset fills that gap.
- **Ground truth (ours)** - averaged phone GPS, EXIF-geolocated photos, six 360° panos, tee-sign cross-checks.

MEASURED, THEN REJECTED

Daylight OSM parquet on S3: planet scan projected past 25 minutes for a 5 km box; the same OSM data via Overpass took 20 seconds. Query kept in the build log.

Terrain Tiles (Terrarium): verified live, not used - the local 1 m DEM renders elevation instead.

Copernicus DEM 30 m: fallback only - about one pixel per 100 ft hole is useless per-hole.

Full provenance, tile IDs, and S3 locations: dataset/sources.md in the repo.

CLOSE · 2:40-2:55

Your course could be next weekend's walk.

Walk it with a phone. Average a point at every tee and basket. Photograph the signs. Shoot the panos.

Run the pipeline. Publish to the same AWS pattern. The map grows hole by hole - honestly, scaffolding and all.



Try it now: bryanchasko.com/cloudcroft-disc-golf/ · the pattern, step by step: /cloudcroft-disc-golf/how-this-works/

APPENDIX A - DO NOT NARRATE UNLESS THE REFRESH SHIPS

backup

Live today vs pending refresh: the label map

ACTION	LIVE TODAY - verified 2026-11-02	PENDING REFRESH - in smoke test
Go to tee box 2	“Stand on the mountaintop — tee box 2”	Mountaintop
Orbit mode	Orbit	Sky-Cam
Walk mode	Walk (WASD)	First Person
Course fly-to	Tiki course · Community course	Byron Ligon · Cloudcroft Community
Immersive entry	Enter first person / VR	VR
Help / judges card	For the judges	? (Help and information)
Pano stepping	← Prev pano · Next pano →	‹ ›
Leave a pano	Back to the walkthrough	Back
Tee box 2 view	Real pad-side field photos (card says no 360° yet)	Jumps straight into the tee-2 360° (config: tee2-area)
Pano hint	(none in the card)	“drag left–right to look around”

Known stale text to confirm in the smoke test: the refresh's static pano-card paragraph still says “No 360° exists at this pad yet…” while its config note jumps straight to the tee-2 pano. Confirm which text actually renders, and narrate only what renders.

Every image and number in this deck

backup

- **Slide 1 / 5 - basket 1 photo:** web/public/evidence/hole1-basket.jpg (Nov 1 field photo, sidecar JSON alongside). Viewed and verified for this deck.
- **Slide 3 - 2D map still:** story/assets/map-hero.png (repo capture, Oct 31). Viewed and verified; labelled with its date on-slide.
- **Slide 4 - tee-2 pano strip + sign:** dataset/panos-web/PXL_20261101_173005849.webp (Pano F; EXIF + distances in dataset/PANO-INDEX.md) and story/assets/hole2-tee-sign.jpg. Sign viewed and read for this deck.
- **Slide 5 - kiosk + Pano A:** web/public/evidence/kiosk.jpg (viewed) and dataset/panos-web/PXL_20261101_170324214.webp.
- **Slide 6 - SW Maps screen:** web/public/figures/fig-swmaps-community-tee2.jpg. Viewed; remark/averaging/elevation read directly from the image.
- **Slide 9 - Tiki aspen pano:** dataset/panos-web/PXL_20261101_180216070.webp (Pano E).
- **Numbers:** 15 photos / 6 panos / 20 screenshots (dataset/survey-assets.json); 21 + 6 survey points and hole facts (README, docs/site-rethink-demo.md, dataset/PANO-INDEX.md); 404.8 ft plays-like check and OSM counts (docs/build-log.md); AWS resources (docs/infra.md, infra-cdk/). No metrics were invented; nothing was downloaded.
- **Honest gaps:** Community holes 3–18 unsurveyed; Tiki points partially identified; Android AR not scaffolded; Location API-key path blocked (Cognito is live). The deck says so wherever it matters.